Research Article Vol. 61, No. 32 / 10 November 2022 / Applied Optics 9415

applied optics

Effect of hologram plane position on particle tracking using digital holographic microscopy

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Received 22 August 2022; revised 12 October 2022; accepted 13 October 2022; posted 17 October 2022; published 4 November 2022

This paper discusses the effect of hologram plane position on the tracking of particle motions in a 3D suspension using digital holography microscopy. We compare two optical configurations where the hologram plane is located either just outside the particle suspension or in the middle of the suspension. In both cases, we record two axially separated holograms using two cameras and subsequently adopt an iterative phase retrieval approach to solve the virtual image problem. We measure the settling motions of 2 μm spheres in a 2 mm thick sample containing 300 to 1500 particles/mm^3. We show that the optical setup where the hologram plane is located in the middle of the sample provides superior tracking results compared to the other, including higher accuracy in the measurement of particle displacement and longer particle trajectories. The accuracy of particle displacement increases by a maximum of 18%, and the trajectory length increases by a maximum of 16%. This superior outcome is due to the less overlapping of the diffraction patterns on the holograms when the separation distance between particles and the hologram plane is minimized. © 2022 Optica Publishing Group

https://doi.org/10.1364/AO.473763

1. INTRODUCTION

Digital holographic microscopy (DHM) has long been used for tracking the movement of micro-particles in three-dimensional (3D) space [1–5], and thus found applications in fluid mechanics [6–14], environment sciences [15–19], and animal locomotion [20–24]. DHM usually involves two steps: digital recording and numerical reconstruction. During the recording, a digital sensor and a microscope objective are used in combination to record a magnified two-dimensional (2D) hologram, which is formed due to the interference between the object and the reference waves. During reconstruction, the recorded hologram is backpropagated to create a 3D optical field, which contains the spatial information of the particles [3]. When the hologram contains a small number of particles, the 3D particle distribution can also be recovered by fitting the hologram to a theoretical model, or the so-called inverse-problem approach [25–27].

When an objective lens with a long working distance is used during recording [3], two optical arrangements are possible as shown in Fig. 1. In the first one, the hologram plane (or the focal plane of the imaging system) is located just outside the sample volume. In the second, the hologram plane is located in the middle of the sample. In most previous DHM implementations, the first optical arrangement is used for the following reason. Reconstructing an inline hologram formed by a single particle will create two images: one real and one virtual, located symmetrically on two sides of the hologram plane [2]. When the hologram plane is located outside the particle suspensions, the virtual image contributes only to out-of-focus noise and can be removed by applying a certain intensity threshold [3]. In contrast, if the hologram plane is located within the sample volume, the virtual image comes into focus during reconstruction, resulting in a failure of the detection of the original 3D particle distribution [28,29].

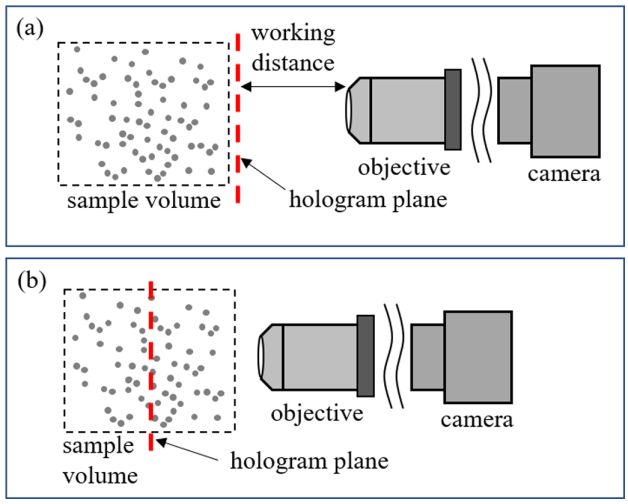


Fig. 1. Two optical arrangements in digital holographic microscopy for three-dimensional particle measurement: (a) hologram plane located outside the sample and (b) hologram plane located within the sample. The hologram plane is the focal plane of the image system.

In a few recent studies [30,31], the second optical configuration has also been experimentally demonstrated to measure the spatial distribution of particles in a 3D particle suspension. In these studies, two cameras were used to simultaneously record two axially separated holograms, and then the Gerchberg–Saxton iterative phase retrieval algorithm [32] was employed to recover the phase information on the hologram planes. A 3D optical field free of virtual images was successfully reconstructed by backpropagating the complex amplitudes (combination of intensity and phase) on the hologram planes. This method of using two axially separated holograms and the Gerchberg–Saxton iterative algorithm to reconstruct twin-image free 3D sample distribution was first introduced and numerically demonstrated by Latychevskaia [33]. The effectiveness of using two holograms and the Gerchberg–Saxton iterative algorithm for reducing virtual image noise has also been demonstrated by other researchers [34,35].

In this paper, we aim to demonstrate which one of the two optical configurations shown in Fig. 1 is best when applying DHM for 3D particle tracking. Following previous studies [30,31], we record two axially separated holograms using two cameras and adopt the Gerchberg–Saxton iterative approach to solve the virtual image problem. We will show that the configuration where the hologram plane is located in the middle of the sample volume provides a superior tracking outcome, including higher accuracy in the measurement of particle displacement and longer particle trajectories. The impact of the separation distance between two hologram planes (D_H) on the reconstruction of particle images has been discussed in our previous studies [28,30]. Virtual images are well suppressed when D_H is sufficiently larger than the diameter of particle d_p. Here, in our experiments, we fixed $D_H = 70d_p$. Although this study and our previous works [28–31] are all related to the application of DHM for 3D particle tracking, this study is different from our previous works. In this study, we examine the effect of hologram plane position on the accuracy of 3D particle tracking, while our previous works focused on the development of new technologies either to distinguish between real and virtual images in inline digital holography [28–30] or improve the axial localization of particles [28,31]. In the following, we describe the experimental methods, results, and finally the conclusion.

2. METHODS

The optical setup of DHM for simultaneously recording two axially separated holograms is illustrated in Fig. 2(a). The light source is a continuous He–Ne laser (Thorlabs, model #HNL100RB, 10 mW) and has a wavelength of $\lambda = 632.8$ nm. The laser beam is attenuated by a neutral density filter, and then filtered by a spatial filter consisting of a 10× objective and a 25 μm pinhole. After passing a convex lens, the beam is collimated into a diameter of 5 mm and illuminates the sample. The sample is located in a glass cuvette (FireFlysci, dimension 2.0 mm × 10 mm × 45 mm) and contains a large number of silver-coated glass spheres (Potters Industry, SG02S40) immersed in de-ionized water. These opaque spheres have a mean diameter $d_p = 2.0$ μm, density $\rho_p = 3.6$ g/m^3, and concentration in the range of $300 < c_p < 1500$ particles/mm^3. Due to gravity, the particles settle down slowly in the glass cuvette. The settling of the particles is recorded by two cameras (FLIR, model #GS3-U3-41C6M-C, 2048 × 2048 pixels, 5.5 μm pixel size). An objective (Edmund #46-143, 5× magnification, infinity corrected, 34 mm working distance) and a tube lens (Edmund #56-863, 2× magnification, 400 mm focal length) are inserted before the cameras to achieve high magnifications. Camera #1 is placed closer to the sample volume than camera #2, so that the two hologram planes (i.e., the two focal planes of the cameras) are separated in the axial direction by a short distance of $D_H = 140$ μm. According to our previous work [28,30], this hologram plane separation distance ($D_H = 70d_p$) is sufficiently large to remove virtual image noise. The calibrated image magnifications are 9.6× and 10.0× for cameras #1 and #2, respectively. The sample volume has a size of 1.2 × 1.2 × 2.00 mm^3.

To compare the two optical configurations shown in Fig. 1, we place the glass cuvette on a translation stage (Thorlabs, model #MT1, precision 25 μm) and carefully move the glass cuvette along the optical axis. We use a Cartesian coordinate system where x and y are the two in-plane directions, and z is the out-of-plane direction opposite to the optical axis. For both optical configurations, we set $z = 0$ at the inner wall of the glass container. The particles are located from $z = 0$ to $z = 2000$ μm. As shown in Fig. 2(b), in the first case, both hologram planes are located outside the sample volume, and hologram plane #1 is located at $z = -50$ μm just outside the particle suspension. In the second case as shown in Fig. 2(c), both hologram planes are located within the sample, and hologram plane #1 is located at $z = 1000$ μm, the middle of the sample volume. For each optical configuration and for each particle concentration tested, we record holograms at 10 frames per second (fps) continuously for 30 s. In total, we have 300 holograms on each camera. Before each recording, the particles are allowed to settle for more than 30 min to minimize the effect of initial disturbance caused by the injection of particles into the container.

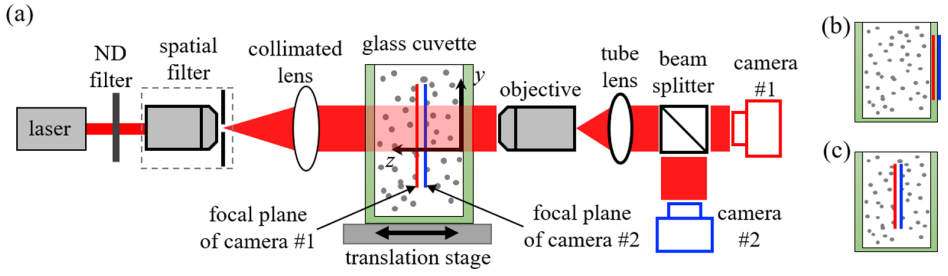


Fig. 2. (a) Optical setup of DHM for simultaneously recording two holograms: (b), (c) optical configurations where the hologram planes are located outside the sample (b) and within the sample (c).

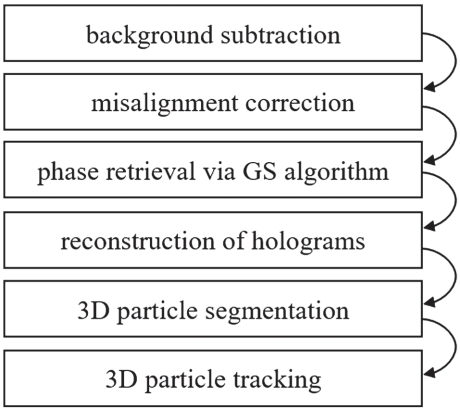


Fig. 3. Data analysis procedure for calculating the particle trajectories from raw holograms.

After recording the data, we follow a data analysis procedure detailed in our previous study [30] for removing the virtual image and calculating the 3D particle distribution. The procedure includes several steps as illustrated in Fig. 3. First, the stationary background noise (e.g., from the optical windows) is removed by subtracting the raw holograms with a background image (obtained by averaging all holograms). Then, the misalignment between the two cameras in the $x-y$ plane is calibrated and corrected by using the following steps. We first create 2D minimal intensity maps, $I_{\min}(x,\ y)$, where each pixel represents the minimum intensity across the entire z range. Consequently, all particles in the 3D volume are projected on $I_{\min}(x,\ y)$. Then, similar to particle image velocimetry, we apply cross-correlations to obtain particle displacements across the two views in x and y directions, and denote them as u and v. Finally, a new hologram I_{H2} is obtained by interpolating the hologram recorded on the second camera, according to $I_{H2}(x,\ y) = I'_{H2}(x+u,\ y+v)$, where $I_{H2}{}'$ is the intensity distribution recorded on the second hologram. We denote the hologram recorded by the first camera as I_{H1}. It should be noted that u and v represent the displacements of each pixel, and could vary at different positions on the image due to rotation between two views or optical aberration. Thus, our approach corrects the misalignment even if there is rotation between two views and optical aberration. After calibration, the images of the same particle on I_{H1} and I_{H2} appear on the same (x, y) position.

Then, the phase distributions on the two hologram planes, denoted as φ_{H1} and φ_{H2}, are estimated by using the Gerchberg–Saxton iterative phase retrieval algorithm [32]. The iterative algorithm involves four steps [34]: (i) propagate the complex amplitude in the first hologram plane to the second one; (2) replace the modulus of the resulting complex amplitude with the measured intensity on the second hologram; (3) back-propagate this new estimate to the first hologram plane; and (4) replace the modulus of the computed complex amplitude with the measured intensity on the first hologram. The algorithm has been demonstrated by many researchers [33–35], and can be applied to both opaque and transparent particles [33]. With the known intensity and phase distributions on the hologram planes, a 3D intensity field denoted as $I(x,\ y,\ z)$ is reconstructed by using the following equation:

$$I(x,\ y,\ z) = |I_{H1}^{0.5}\exp(\varphi_{H1}) \otimes h(x,\ y,\ z)|^2, \tag{1}$$

where $h = z\exp\{ik(x^2+y^2+z^2)^{0.5}\}/\ [i\lambda(x^2+y^2+z^2)]$ is the Rayleigh–Sommerfeld diffraction kernel, $k = 2\pi/\lambda$, and $\otimes$ represents a convolution. It should be emphasized that $I(x,\ y,\ z)$ calculated from Eq. (1) is free of virtual images. This method is distinct from single-hologram-based DHM, where the reconstruction is based solely on the intensity on the hologram plane, and the resulting 3D intensity field is influenced by the virtual image.

Once the 3D intensity field is obtained, the 3D particle distribution is detected by applying 3D image segmentation technology, demonstrated in our previous study [29,31]. A global threshold is manually selected based on the intensities of the in-focus particles, and is applied to distinguish between the background and the particle features. Although a local threshold could be used to improve particle detection [3], here, we use the global threshold for the purpose of comparing the two optical configurations. The particle position is calculated based on the intensity-weighted centroid of the 3D particle trace. Finally, the particle movement is tracked via a particle tracking algorithm developed in [36]. This algorithm has been demonstrated in our previous studies for tracking the 3D Brownian motions of micro-spheres using DHM [29,31].

3. RESULTS AND DISCUSSION

Figure 4 shows the intensity distributions on the hologram planes (I_{H1}) at three different particle concentrations: $c_p = 330$, 800, and 1400 particles/mm^3. The top row is obtained when the hologram planes are located outside the particle suspensions, and the bottom row corresponds to the case where the hologram planes are located in the middle of the particle suspensions. As expected, with increasing the particle concentration, the diffraction patterns of individual particles are more likely to be overlapped with each other. However, for cases where the hologram planes are located in the middle of the sample, the diffraction patterns are less overlapped and more distinguishable. The reason is that placing the hologram planes in the middle of the sample effectively reduces the separation distance between particles and the hologram plane, and thus the signal generated by each particle is more localized on the image.

Before comparing the two optical configurations, we first demonstrate that the virtual image is successfully removed in the 3D reconstruction by the use of Eq. (1). Figures 5(a) and 5(b) show the two holograms recorded by cameras #1 and #2, respectively. Only a small 200 × 200 pixel region of the original holograms is shown. A sample particle is located in the center of the shown region. Figures 5(c) and 5(d) show the intensity distribution in the $y-z$ plane at the center of the sample particle obtained by reconstructing the holograms. In Fig. 5(c), the reconstruction is based on the use of one hologram, and as a result, both real and virtual images present on the two sides of the hologram plane. In contrast, in Fig. 5(d), the reconstruction is based on the use of two holograms and Eq. (1), and consequently the virtual image is completely removed. Comparing the intensity distributions in the $x-y$ plane [Figs. 5(e) and 5(f), obtained by projecting the minimal intensity across the entire

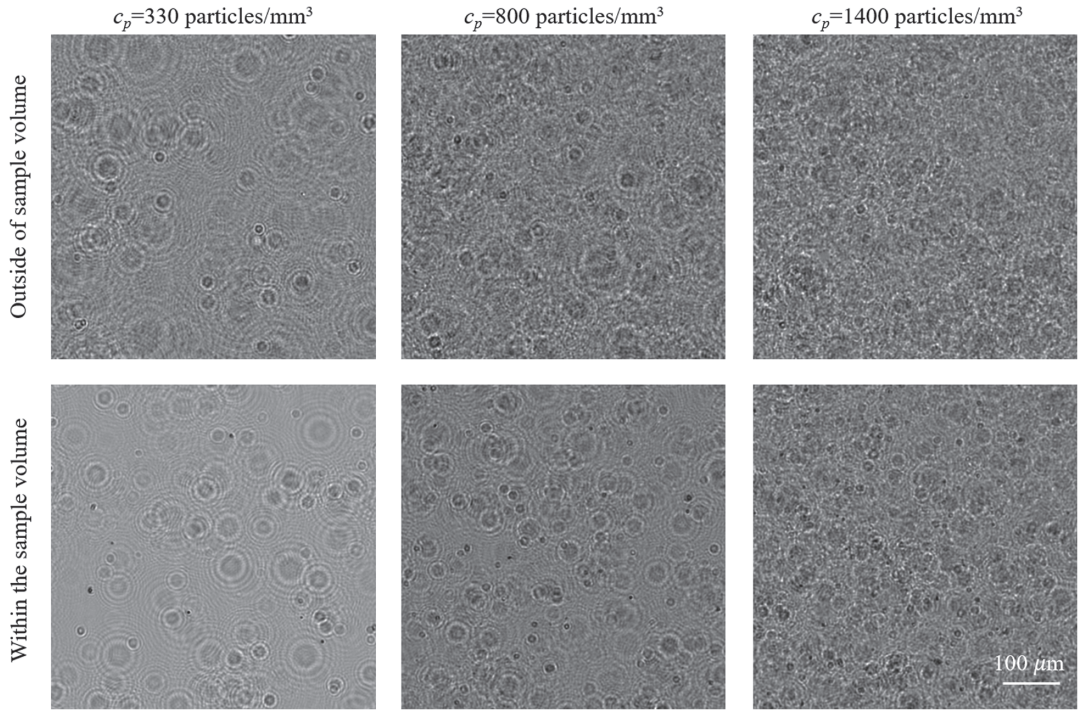


Fig. 4. Intensity distributions on the hologram plane obtained at three different particle concentrations: $c_p = 330, 800, 1400$ particles/mm^3. The top and bottom rows are obtained when the hologram plane is placed outside the particle suspension, and in the middle of the particle suspension, respectively. The scale bar in the last image applies to all others. All six holograms are shown in the same intensity scale.

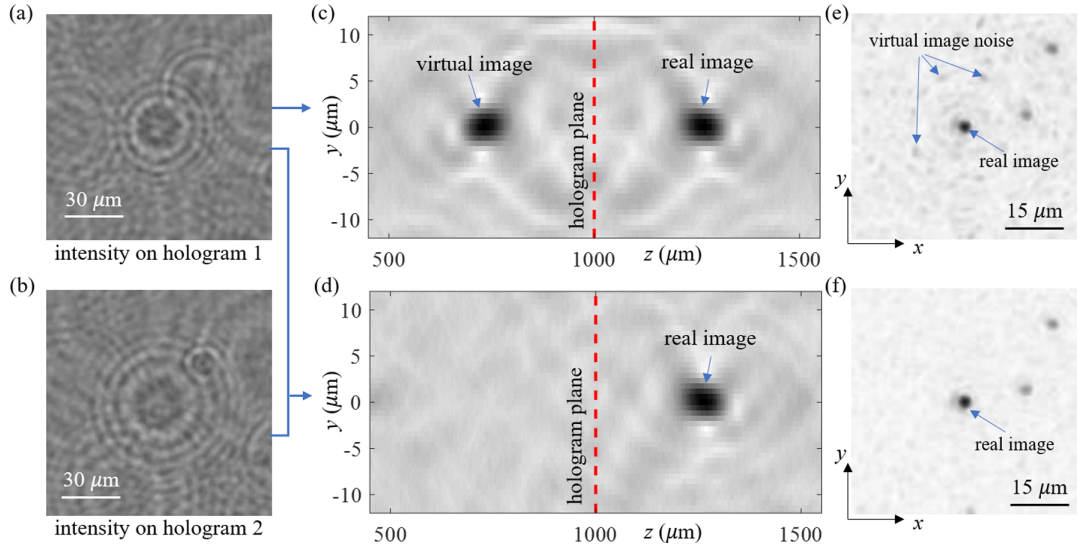


Fig. 5. (a), (b) Sample holograms recorded by the two cameras; (c), (d) reconstructed intensity distributions in the $y-z$ plane in the center of a particle; and (e), (f) minimal intensity projected to $x-y$ plane. (c), (e) Obtained by reconstructing the intensity on hologram plane #1 and (d), (f) obtained by reconstructing the complex amplitude on hologram plane #1 estimated by propagating the light between the two holograms. Hologram plane #1 is located at $z = 1000$ μm.

depth to the $x-y$ plane], the out-of-focus virtual image noise is clearly suppressed by using two axially separated holograms.

Figures 6(a) and 6(b) show the calculated 3D particle trajectories obtained at a particle concentration of $c_p = 330$ particles/mm^3. In Fig. 6(a), the two hologram planes are located outside the sample volume, and in Fig. 6(b), the two hologram planes are located in the middle of the sample. Clearly, both optical configurations resolve the settling motion of the particles in the gravitational direction (i.e., y direction). The particle displacements in x and z directions are much smaller than the displacement in the y direction.

To quantify which optical configuration provides a better tracking result, we first compare the particle displacements in x and z directions at each time step, denoted as dx and dz.

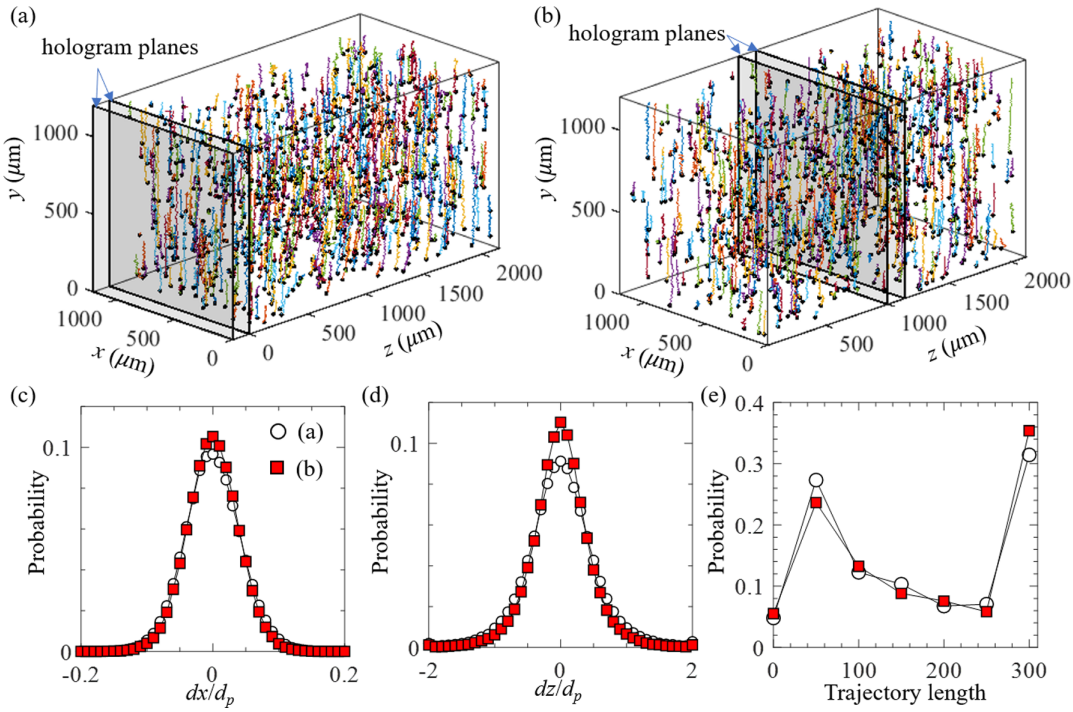


Fig. 6. (a), (b) Particle trajectories obtained from the two optical configurations. Dark dots indicate the end of each trajectory. (c)–(e) Probability distributions of particle displacement in x direction (c), particle displacement in z direction (d), and particle trajectory length (e). The data are obtained at a particle concentration of 330 particles/mm^3. Results at higher particle concentrations show similar trends.

Figures 6(c) and 6(d) show the probability distributions of dx and dz, respectively, obtained from the two optical configurations. Ideally, dx and dz should be equal to zero since the particle has no motions in x or z direction. But in reality, dx and dz have Gaussian distributions centered at zero due to the uncertainty in the localization of particle position. The uncertainty of dx is related to the in-plane resolution of the imaging system. The uncertainty of dz is much larger than the uncertainty of dx, and is due to the elongation of a particle image in the z direction, or the so-called depth-of-focus issue [2]. Nevertheless, if the particle concentration is low, the uncertainty in z direction could be significantly improved by an inverse problem approach [26,27]. In this paper, we discuss only applications where the particle concentration is high.

As shown in Figs. 6(c) and 6(d), when comparing the two optical configurations, the one where the hologram planes are located in the middle of the sample has smaller dx and dz, indicating higher accuracy in the measurement of particle displacement. Specifically, by placing the hologram planes in the middle of the sample, the standard deviation of dx reduces from 0.084 to 0.078 μm, and the standard deviation of dz reduces from 1.07 to 0.91 μm. This result agrees with the more distinguishable diffraction patterns recorded on the holograms when placing the hologram planes in the middle of the sample.

Furthermore, we compare the length of the particle trajectory to verify which optical configuration provides a better tracking result. A trajectory of the same particle can sometime be broken into multiple shorter ones due to the misdetection of the particle in the time series of holograms. The misdetection is usually caused by the low signal-to-noise ratio of the reconstructed particle image. Figure 6(e) shows the probability distribution of the length of the particle trajectory for the two optical configurations. The trajectory length is defined as the number of frames in which a same particle is continuously detected and tracked. In each dataset, we recorded 300 frames on each camera. Thus, a trajectory length of 300 means that the particle is successfully detected and tracked across all frames. Clearly, the optical setup where the hologram planes are located in the middle of the sample has a higher probability to obtain longer particle trajectories. Specifically, by placing the hologram planes in the middle of the sample, the average trajectory length increases from 163 to 170. Again, the result confirms that placing the hologram planes in the middle of the sample provides a better tracking result.

Finally, we perform a comparison of the two optical arrangements at other particle concentrations in the range of 300 to 1500 particles/mm^3. Figures 7(a) and 7(b) show the standard derivations of particle displacements in x and z directions, denoted as σ_{dx} and σ_{dz}. Figure 7(c) shows the average length of the particle trajectories. As expected, with increasing the particle concentration, both the accuracy of particle displacement and the length of particle trajectory reduce, indicating a degradation of the quality of the tracking result. This trend is inconsistent with previous studies that show the limitation of digital holography at high particle seeding density [30,37]. Interestingly, when comparing the two optical configurations, we find that the one where the hologram planes are located in the middle of the sample provides higher accuracy in the measurement of particle displacements [Figs. 7(a) and 7(b)], as well as longer particle trajectories [Fig. 7(c)]. Varying the particle concentration from 300 to 1500 particles/mm^3, the improvement of particle localization in x direction varies from 5% to 17%,

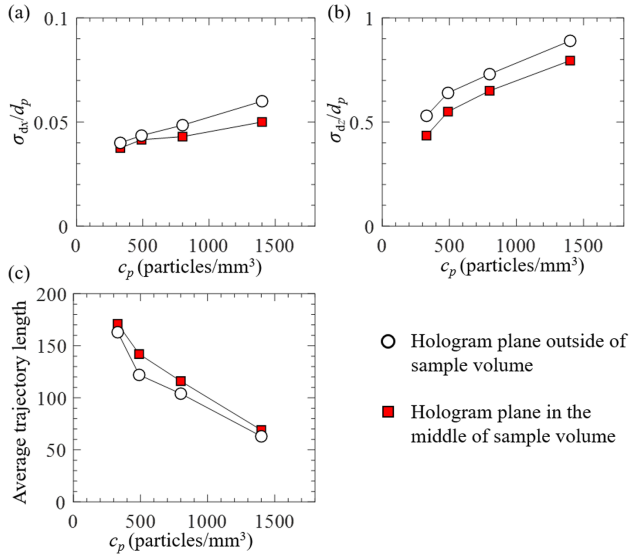


Fig. 7. (a), (b) Standard derivations of particle displacement in (a) x and (b) z directions; and (c) average particle trajectory length. The trajectory length is defined as the number of frames where a particle is continuously detected and tracked. Here, the total number of frames is 300, and the trajectory length does not exceed 300.

the improvement of particle localization in z direction changes from 11% to 18%, and the improvement of particle trajectory length changes from 5% to 16%. Again, the superior outcome is because the diffraction patterns recorded on the holograms are much clearer when the hologram planes are closer to the sample.

When the holograms planes are located in the middle of the sample, some particles are inevitably close to the focus plane. To test if our reconstruction algorithm could resolve these particles, we performed the following numerical simulations. First, holograms of an opaque particle with a diameter $d_p = 2\ \mu$m are created based on Mie-scattering theory [38]. For every particle, two holograms are generated, similar to the experiments. Hologram plane #1 is located at $z = 0$, hologram plane #2 is located at $z = -10d_p$, and the particle is located at $z = z_p$. Thus, the distance between the particle and hologram plane #1 is z_p. Then, we use the two axially separated holograms and Gerchberg–Saxton iterative algorithm to reconstruct a 3D intensity field according to Eq. (1). Figures 8(a)–8(c) show a sample hologram and the corresponding reconstructed intensity distributions at the center of the particle. Clearly, the particle comes to focus at $z = z_p$, and the virtual image noise is well suppressed.

To study the impact of z_p on the quality of particle reconstruction, we vary z_p from $2d_p$ to $40d_p$, and calculate a focal parameter defined as $F(z) = I_s(z) - I_p(z)$. Here, I_s and I_p are the mean intensities of a small area, $5\times$ and $1\times$ the particle diameter, respectively, surrounding the particle. Ideally, this focal parameter reaches a peak at $z = z_p$, and the particle axial position can be detected based on the peak position. Figure 8(d) shows sample focal parameters for particles located at $z_p = 2d_p$, $10d_p$, and $25d_p$. All profiles show a clear peak around $z = z_p$. Figure 8(e) shows the error of particle axial position as a function of z_p (i.e., the distance between the particle and hologram plane #1). The error has a maximum of $3d_p$ when the particle is located very close to the hologram plane ($z_p < 3d_p$), and reduces to $1d_p$ when z_p is larger than $8d_p$ and to zero when z_p is larger than $15d_p$.

4. CONCLUSION

When applying the DHM for 3D particle tracking, the hologram plane is typically located outside the sample so that the

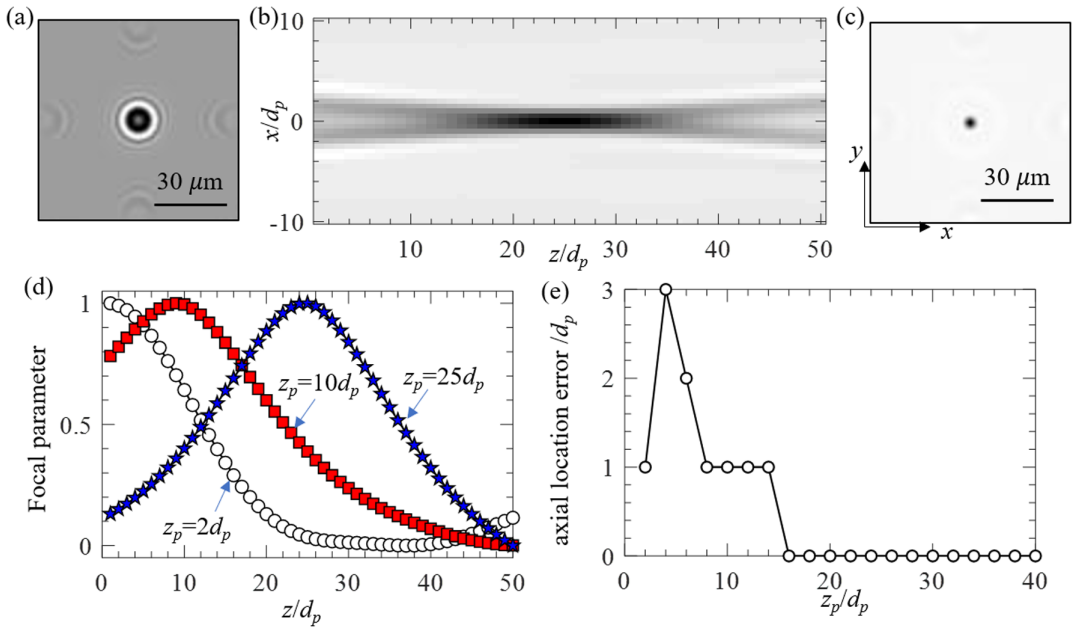


Fig. 8. (a) Sample hologram of a particle located at $z_p = 25d_p$ generated based on Mie-scattering theory; (b), (c) intensity distributions at the center of the particle in $x-z$ and $x-y$ planes, respectively, obtained by using two holograms and Eq. (1); (d) sample focal parameters for particles located at $z_p = 2d_p$, $10d_p$, and $25d_p$; (e) axial localization error as a function of distance between particle and hologram plane #1. The parameters in the simulation are: $\lambda = 633$ nm, pixel size 1 μm, image size 200 μm, $D_H = 20$ μm, and $d_p = 2$ μm.

virtual image does not come into focus during reconstruction. Recently, the use of two axially separated holograms and the Gerchberg–Saxton iterative algorithm has solved the virtual image problem, and thus allowed the implementation of a new optical arrangement where the hologram plane is located in the middle of the sample. In this work, we discuss the effect of hologram plane position on the tracking of particle motions in 3D suspensions. We demonstrate that the optical arrangement where the hologram plane is located in the middle of the particle suspension provides superior tracking results compared to the one where the hologram plane is located outside the sample. The reason for the superior outcome is due to the less overlapping of the diffraction patterns on the holograms when the separation distance between particles and the hologram plane is minimized. Therefore, when the virtual image can be removed, we suggest the usage of the optical arrangement where the hologram plane is located in the middle of the sample for 3D particle tracking. Note that this study tested only opaque particles. Further study is required to test whether our conclusion could apply to transparent micro-spheres that cause strong phase shifts. The conclusion is limited to cases where the particle concentration is high and the extent of overlapping among diffraction patterns is large. At low particle concentrations, however, placing the hologram plane outside the sample may provide better results since the distance between the hologram and particle is larger and the information is encoded on a larger area.

Funding. Office of Naval Research (N00014-20-1-2170).

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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